



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

FULL SYLLABUS

14-04-2024

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	D	B	B		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,C,D	C,D	A,B,D	B,D	A,B,D	A,B,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	5.66 to 5.67	5.00	1.00	3905.00 to 3915.00	10.00	150.00
SECTION-II (ii)	Q.	7	8	9			
	A.	3	5	126			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	D	A	B	B		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	C,D	A,B,C,D	A,B,D	B,C	A,B,C	A,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	7.40	0.02	3.00	4.00	4.00	16.00
SECTION-II (ii)	Q.	7	8	9			
	A.	202	2	60			

PART-3 : MATHEMATICS

SECTION-I(i)	Q.	1	2	3	4		
	A.	A	B	B	C		
SECTION-I(ii)	Q.	5	6	7	8	9	10
	A.	B,D	B,C	A,B,C	A,D	B,D	A,B
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.00	8.00	5.00	6.00	24.00	40.00
SECTION-II (ii)	Q.	7	8	9			
	A.	9	8	9			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (C)

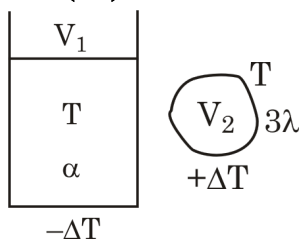
Suppose that the two conductors carry free charge Q and $-Q$. Consider a closed surface enclosing the conductor with charge Q . Using ohm's and Gauss's Law,

$$I = \oint \vec{J} \cdot d\vec{S} = \oint \sigma \vec{E} \cdot d\vec{S} = \sigma \oint \vec{E} \cdot d\vec{S}$$

$$I = \sigma \frac{Q}{\epsilon_0 \epsilon_r} \Rightarrow \frac{V}{R} = \sigma \frac{Q}{\epsilon_0 \epsilon_r}$$

$$C = \frac{Q}{V} = \frac{\epsilon_0 \epsilon_r}{\sigma R}$$

2. Ans (D)



$$V_1(1 - \alpha \Delta T) + V_2(1 + 3\lambda \Delta T) = V_1$$

$$-V_1 \alpha \Delta T + V_2 + 3\lambda V_2 \Delta T = 0$$

$$\Delta T = \frac{V_2}{\alpha V_1 - 3\lambda V_2}$$

3. Ans (B)

Object at distance $2f$ produce real image of same size and inverted.

Object lying between f and $2f$ produce real inverted magnified image.

Object lying outside $2f$ produce real inverted diminished image.

4. Ans (B)

$$\frac{dn}{n} = \frac{1}{100}, dn = \frac{2}{100}$$

$$R = \left(\frac{n-1}{n+1} \right)^2$$

$$\ell n R = 2 \ell n (n-1) - 2 \ell n (n+1)$$

$$\frac{1}{R} dR = \frac{2}{n-1} dn + \frac{2}{n+1} dn$$

$$= \frac{2}{1} \times \frac{2}{100} + \frac{2}{3} \times \frac{2}{100} = \frac{16}{300}$$

$$\frac{dR}{R} \% = \frac{16}{3} \%$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (A,B,C,D)

$$p_t = \frac{2z_1 z_2}{z_1 + z_2} p_i$$

$$p_r = \frac{z_2 - z_1}{z_1 + z_2} p_i$$

$$z = \rho v = \frac{B}{v}$$

6. Ans (C,D)

$$2P_0 V_0 T_0 \quad P_0 V_0 T_0$$

$$2P_0 V T \quad V_0 2P_0 T$$

$$\frac{2P_0 V_0}{T_0} + \frac{P_0 V_0}{T_0} = \frac{2P_0(V + V_0)}{T}$$

$$\frac{V}{T} + \frac{V_0}{T} = \frac{V_0}{T_0} + \frac{V_0}{2T_0} = \frac{3}{2} \frac{V_0}{T_0} \quad \dots(1)$$

$$U_i = \frac{2P_0 V_0}{RT_0} \times \frac{3}{2} RT_0 + \frac{P_0 V_0}{RT_0} \times \frac{3}{2} RT_0$$

$$U_f = \frac{2P_0(V_0 + V)}{RT} \times \frac{3}{2} RT = 3P_0(V_0 + V)$$

$$-\Delta W = \Delta U$$

$$2P_0(V_0 - V) = 3P_0(V_0 + V) - \frac{9}{2} P_0 V_0$$

$$4V_0 - 4V = 6V_0 + 6V - 9V_0$$

$$7V_0 = 10V$$

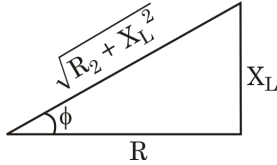
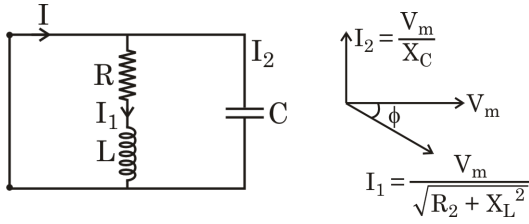
$$V = \frac{7V_0}{10}$$

$$\ln(1)$$

$$\frac{7V_0}{10T} + \frac{V_0}{T} = \frac{3}{2} \frac{V_0}{T_0}$$

$$T = \frac{17T_0}{15}$$

7. Ans (A,B,D)



For resonance, $I_2 = I_1 \sin \phi$

$$\frac{V_m}{X_C} = \frac{V_m}{\sqrt{R^2 + X_L^2}} \times \frac{X_L}{\sqrt{R^2 + X_L^2}}$$

$$\Rightarrow R^2 + X_L^2 = X_L \cdot X_C$$

$$\Rightarrow R^2 + \omega^2 L^2 = \frac{L}{C}$$

$$\Rightarrow \omega_r = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

8. Ans (B,D)

$$\Delta\theta = \frac{2\lambda}{a} \text{ (angular width of central maximum).}$$

$$\frac{2\lambda}{a} - \frac{2\lambda}{a} = 0.3 \times \frac{2\lambda}{a}$$

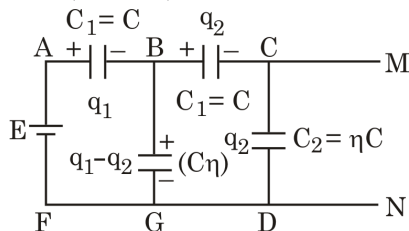
$$\lambda' = \lambda - 0.3\lambda = 4200 \text{ \AA}$$

$$\text{In case (2)} \Delta\theta = \frac{2\lambda}{\mu a}$$

$$\frac{2\lambda}{a} - \frac{2\lambda}{\mu a} = 0.3 \times \frac{2\lambda}{a} \Rightarrow 1 - \frac{1}{\mu} = 0.3$$

$$\Rightarrow \mu = 1.43$$

9. Ans (A,B,D)

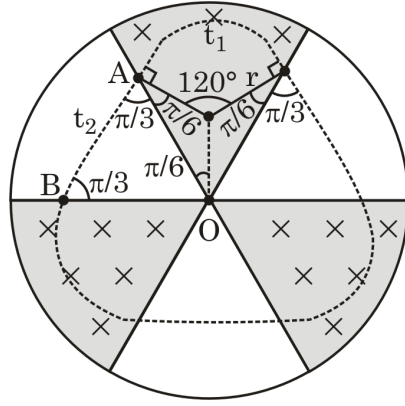


$$\text{In loop ABGFA} \quad \frac{q_1}{C} + \frac{q_1 - q_2}{\eta C} = E$$

$$\text{In loop ABCDGFA} \quad \frac{q_1}{C} + \frac{q_2}{C} + \frac{q_2}{\eta C} = E$$

$$V_{mn} = \frac{q_2}{\eta C} = \frac{E}{n^2 + 3n + 1}$$

10. Ans (A,B,D)



$$r = \frac{mv}{qB} \text{ -- (A)}$$

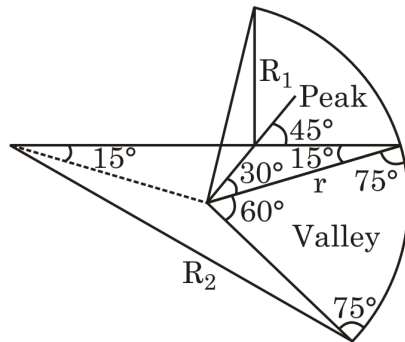
$$t_1 = \frac{2\pi m}{3qB}$$

$$AB = AO = 2r \cos 30 = r\sqrt{3} = \frac{mv}{qB} \sqrt{3}$$

$$t_2 = \frac{AB}{v} = \frac{m}{qB} \sqrt{3}$$

$$T = 3(t_1 + t_2) = 3 \left(\frac{2\pi m}{3qB} + \frac{\sqrt{3}m}{qB} \right)$$

$$= \frac{m}{qB} (2\pi + 3\sqrt{3}) \text{ -- (B)}$$



$$\frac{r}{\sin 135^\circ} = \frac{R_1}{\sin 30^\circ}$$

$$R = \frac{r \sin 30^\circ}{\sin 135^\circ} = r \frac{1}{2} \times \sqrt{2} = \frac{r}{\sqrt{2}}$$

$$\frac{r}{\sin 15^\circ} = \frac{R_2}{\sin 150^\circ}$$

$$R_2 = \frac{r \sin 150^\circ}{\sin 15^\circ} = \frac{r}{2} \frac{2\sqrt{2}}{\sqrt{3} - 1} = \frac{r\sqrt{2}}{\sqrt{3} - 1}$$

$$\frac{R_1}{R_2} = \frac{B'}{B} = \frac{r}{V_2} \frac{\sqrt{3} - 1}{\sqrt{2}} = \frac{\sqrt{3} - 1}{2}$$

PART-1 : PHYSICS

SECTION-II (i)

2. Ans (5.00)

$$B = \frac{\mu_0 4i}{\sqrt{2}\pi}$$

$$\phi_{\text{one turn of coil}} = \frac{4i}{\sqrt{2}\pi} \times \frac{1}{20}$$

$$\therefore \text{induced emf} = -5 \times \frac{1}{20} \frac{4\mu_0}{\sqrt{2}\pi} \times \frac{di}{dt}$$

$$= -2\sqrt{2} \frac{di}{dt} \times 10^{-7}$$

$$= 2\sqrt{2} \times 10^{-7} \times 20 \frac{d \sin(1000 t)}{dt}$$

$$= 4\sqrt{2} \times 10^{-3} \cos 1000 t$$

$$\therefore \text{Peak value of emf} = 4\sqrt{2} \text{ m volt}$$

$$= 4 \times 1.414$$

$$= 5.656 \approx 5.66 \text{ mV}$$

For part -2

Induced emf

$$= 2\sqrt{2} \times 10^{-7} \left[20 \frac{d \sin 1000t}{dt} + 5 \frac{d \sin(3000t)}{dt} \right]$$

$$= 2\sqrt{2} \times 10^{-3} [2 \cos 1000t + 1.5 \cos 3000t]$$

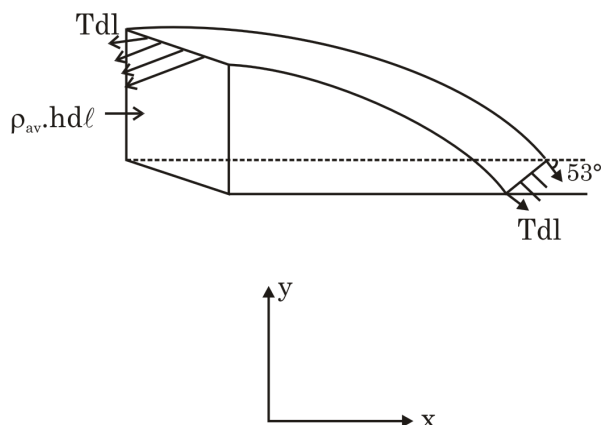
$$= \sqrt{2} [4 \cos 1000 t + 3 \cos 3000t]$$

$$\therefore e_{\text{rms}} = \sqrt{\frac{(4\sqrt{2})^2 + (3\sqrt{2})^2}{2}}$$

$$= \sqrt{(16 + 9)} = 5 \text{ m volt}$$

4. Ans (3905.00 to 3915.00)

Consider an element of liquid column



$$\sum F_x = 0 \quad T dl (1 - \cos 53) = \frac{\rho g h}{2} \cdot h dl$$

$$F_b \uparrow \quad 2T \ell \sin 53^\circ$$

$$mg \downarrow$$

$$\frac{1}{2} \left(1 - \frac{3}{5} \right) = \frac{4 \times 10^3 \times 10}{2} h^2$$

$$\frac{2}{400 \times 5 \times 10^4} = h^2$$

$$h = \sqrt{\frac{1}{10^6}} = 1 \text{ mm}$$

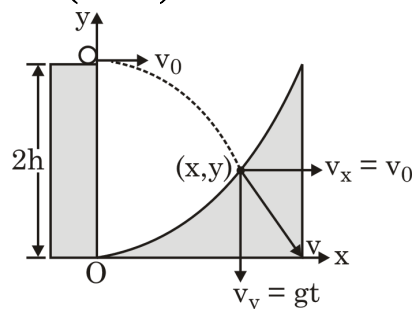
Now for equilibrium of cylinder

$$F_b + 2T \ell \cdot \frac{4}{5} = \rho_{\text{cy}} \cdot g \pi R^2 \ell$$

$$F_b = \rho_{\text{liq}} g (h + R) \times 2R \ell - \left(R \times 2R \times \ell - \frac{\pi R^2 \ell}{2} \right) \rho g$$

$$\rho_{\text{cy}} = 3909.82 \text{ kg/m}$$

5. Ans (10.00)



$$x = v_0 t$$

$$2h - y = \frac{1}{2} \cdot gt^2$$

$$y = 2h - \frac{1}{2}gt^2$$

$$y = \frac{x^2}{2h} = \frac{v_0^2 t^2}{2h}$$

$$2h - \frac{1}{2}gt^2 = \frac{v_0^2 t^2}{2h}$$

$$v_0^2 t^2 = 4h^2 - ght^2$$

$$t^2 = \frac{4h^2}{v_0^2 + gh}$$

$$t = \frac{2h}{\sqrt{v_0^2 + gh}}$$

$$v^2 = v_0^2 + g^2 t^2$$

$$= v_0^2 + g^2 \cdot \frac{4h^2}{v_0^2 + gh} = v_0^2 + \frac{4g^2 h^2}{v_0^2 + gh}$$

$$K \rightarrow \text{minimum, } v^2 \rightarrow \text{minimum}$$

$$\frac{dv^2}{dv_0^2} = 0$$

$$1 - \frac{4g^2 h^2}{(v_0^2 + gh)^2} = 0$$

$$(v_0^2 + gh)^2 = 4g^2 h^2$$

$$v_0^2 + gh = \pm 2gh$$

$$v_0 = gh$$

$$v_0 = \sqrt{gh} = \sqrt{10 \times 10} = 10 \text{ m/s}$$

$$v^2 = 100 + \frac{4 \times 100 \times 100}{100 + 100} = 300 \text{ m/s}$$

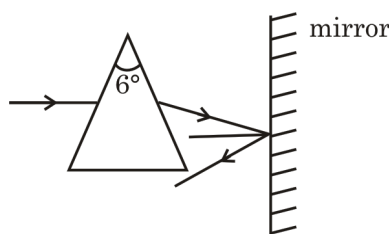
$$K = \frac{1}{2} \times 1 \times 300 = 150 \text{ J}$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (3)

$$\delta = (\mu - 1)A = (2 - 1) \times 6^\circ = 6^\circ$$



If we rotate plane mirror by 3° in clock wise direction then reflected ray is rotated by 6° in clockwise direction.

8. Ans (5)

$$f_0 = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$f' = \frac{1}{2\pi} \sqrt{\frac{K}{m} - \frac{b^2}{4m^2}}$$

$$f_0 = \frac{2}{\sqrt{3}} f' \Rightarrow \frac{K}{m} = \frac{4}{3} \left(\frac{K}{m} - \frac{b^2}{4m^2} \right)$$

$$\frac{b^2}{4m^2} = \frac{1}{4} \cdot \frac{K}{m} \Rightarrow b^2 = 25 \Rightarrow b = 5$$

9. Ans (126)

$$\Delta U = 0$$

$$\Delta U = \Delta U_{\text{lead}} + \Delta U_{\text{water}}$$

$$\Delta Q_{\text{Lead}} = \Delta U_{\text{Lead}} = m_{\text{Lead}} \cdot C_P \text{ Lead } (T_f - T_i)$$

$$0 = 0.6 \times C_P \times (292.8 - 373.0) + 0.2 \times 386 (292.8 - 290.0)$$

$$+ 0.5 \times 4180 (292.8 - 290.0)$$

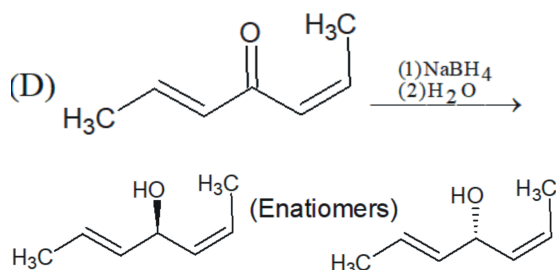
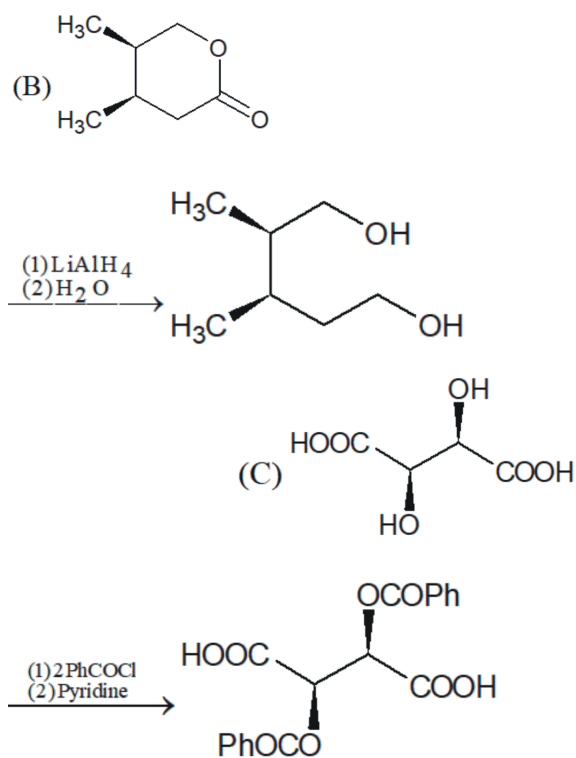
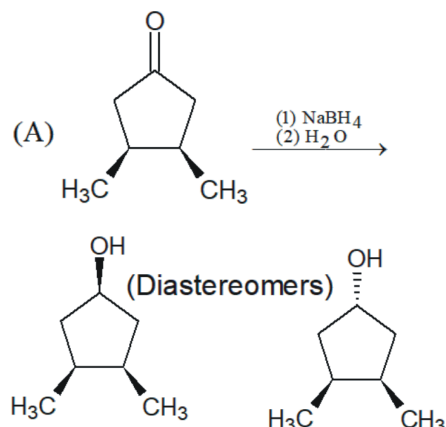
Which in

$$C_P = 126 \text{ J kg } ^\circ\text{K}^{-1}$$

PART-2 : CHEMISTRY

SECTION-I (i)

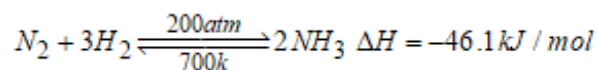
2. Ans (A)



3. Ans (B)

In Haber's process iron oxide is used as a catalyst with small amounts of K_2O & Al_2O_3 to increase the rate of attainment of equilibrium.

Process $\rightarrow$



4. Ans (B)

(i) If hybridisation is dsp^2 then shape will be square planer.

(ii) In case of 2 lone pair & 4 bond pair (sp^3d^2) shape will be square planer.

PART-2 : CHEMISTRY

SECTION-I (ii)

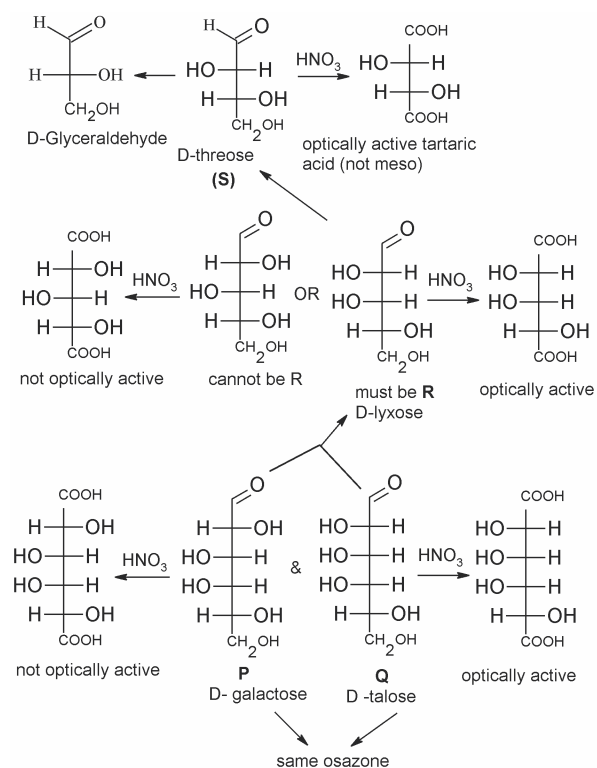
6. Ans (A,B,C,D)

$$\lambda = 2\pi a_0 \left(\frac{n}{Z} \right)$$

$$\text{for He}^+ \text{ ion, } n = 2, Z = 2, \lambda_2 = 2\pi a_0 \left(\frac{2}{2} \right) = X$$

$$\Rightarrow 2\pi a_0 = X$$

7. Ans (A,B,D)



8. Ans (B,C)

Compound(P) and Compound(Q) can be differentiated by **NOT** by aq.NaOH. Only compound(R) gives foul smell of isocyanide with CHCl_3 -KOH

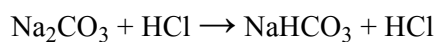
9. Ans (A,B,C)

(d) is incorrect, Hg is not a transition elements.

PART-2 : CHEMISTRY

SECTION-II (i)

1. Ans (7.40)

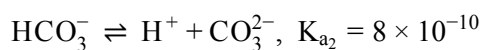


0.1M

$$[\text{H}^+] = \sqrt{K_{a1} K_{a2}} = \sqrt{16 \times 10^{-16}} = 4 \times 10^{-8}$$

$$\text{pH} = 8 - \log 4 = 8 - 0.6 = 7.4$$

2. Ans (0.02)

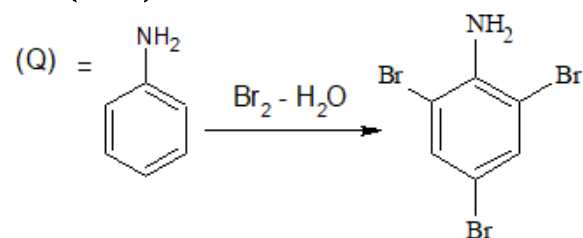


$$0.1 \text{ M} \quad 4 \times 10^{-8} \text{ M}$$

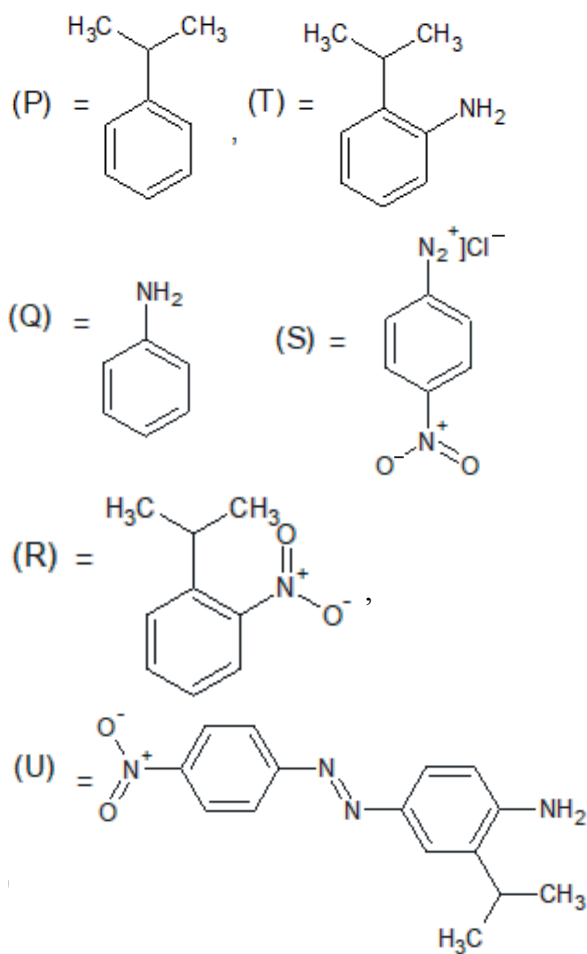
$$8 \times 10^{-10} = \frac{(4 \times 10^{-8}) [\text{CO}_3^{2-}]}{0.1}$$

$$\Rightarrow [\text{CO}_3^{2-}] = 2 \times 10^{-3} \text{ M}$$

3. Ans (3.00)



4. Ans (4.00)



6. Ans (16.00)

Let initial concentration of A = $[\text{A}_0]$

$$\text{Rate} = -\frac{1}{3} \frac{d[\text{A}]}{dt} = k[\text{A}]^n;$$

$$-\int_{[\text{A}_0]}^{[\text{A}_t]} \frac{d[\text{A}]}{[\text{A}]^n} = 3k \int_0^t dt; -\left[\frac{[\text{A}]^{-n+1}}{-n+1} \right]_{[\text{A}_0]}^{[\text{A}_t]} = 3kt$$

$$\left(\frac{1}{[\text{A}_t]^{n-1}} - \frac{1}{[\text{A}_0]^{n-1}} \right) = (n-1) 3kt;$$

$$\frac{1}{[\text{A}_t]^{n-1}} = (n-1) 3kt + \frac{1}{[\text{A}_0]^{n-1}}$$

According to graph $n-1=3$, $n=4$, slope = $\tan 45^\circ$

$$= 1 = (n-1) 3k, k = 1/9$$

$$\text{Rate} = k$$

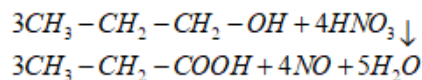
$$[\text{A}]^4 = \frac{1}{9} (0.2)^4 = \frac{16}{9} \times 10^{-4} \text{ M min}^{-1}$$

$$P = \frac{16}{9} \text{ so } 9P = 16$$

PART-2 : CHEMISTRY

SECTION-II (ii)

7. **Ans (202)**



$$\frac{n_{HNO_3}}{n_{Alcohol}} = \frac{4}{3}$$

$$n_{HNO_3}(\text{Expected}) = \frac{4}{3} \times \frac{180}{60} = 4$$

as % reaction yield is 80%

Hence

$$n_{HNO_3}(\text{actual}) = 4 \times 0.8 = 3.2$$

$$w_{HNO_3} = 3.2 \times 63 = 201.6$$

8. **Ans (2)**

(2) & (6) are not used for softening of water

9. **Ans (60)**

$$\frac{d}{P} = 2 + 0.02 P$$

$$\text{at } P \rightarrow 0 \Rightarrow \lim_{P \rightarrow 0} \left(\frac{d}{P} \right) = 2 = \frac{M}{RT}$$

$$\Rightarrow M = 2RT = 60 \text{ gm/mole}$$

PART-3 : MATHEMATICS

SECTION-I(i)

1. **Ans (A)**

$$\frac{\alpha^{-\alpha\beta} [\alpha^\alpha \alpha^\beta - 4\alpha^{-\alpha\beta}] + \beta^{-\alpha\beta} [\beta^\alpha \beta^\beta - 4\beta^{-\alpha\beta}]}{\alpha^{\alpha+\beta-\alpha\beta} + \beta^{\alpha+\beta-\alpha\beta} - 4(\alpha^{-2\alpha\beta} + \beta^{-2\alpha\beta})}$$

$$\frac{\alpha^{\alpha+\beta} + \beta^{\alpha+\beta}}{\alpha^{\alpha+\beta+1} + \beta^{\alpha+\beta+1}}$$

$$\alpha + \beta = 4, \alpha\beta = -3$$

$$\frac{\alpha^7 + \beta^7 - 4(\alpha^6 + \beta^6)}{\alpha^5 + \beta^5}$$

$$S_n = \alpha^n + \beta^n$$

$$S_n - 4S_{n-1} - 3S_{n-2} = 0$$

$$S_7 - 4S_6 - 3S_5 = 0$$

2. **Ans (B)**

Since $a_{n+2} = 1 - \frac{1}{a_{n+1}} = 1 - \frac{a_n}{a_n - 1} = \frac{-1}{a_n - 1}$, it follows that

$$a_{n+3} = 1 - \frac{1}{a_{n+2}} = 1 + (a_n - 1) = a_n,$$

so is a periodic sequence with a period 3. Since

$$a_1 = 2, a_2 = \frac{1}{2}, a_3 = -1, \text{ so } P_3 = -1 \quad \text{and}$$

$$P_{2017} = (P_3)^{672} \cdot P_1 = 2$$

3. **Ans (B)**

$$\tan 3^\circ + 2 \tan 6^\circ + 4 \tan 12^\circ + \frac{8 \tan 24^\circ}{8(1 - \tan^2 12^\circ)}$$

$$\tan 3^\circ + 2 \tan 6^\circ + 4 \tan 12^\circ + \frac{4}{2 \tan 12^\circ}$$

$$\tan 3^\circ + 2 \tan 6^\circ + \frac{2(1 - \tan^2 6^\circ)}{\tan 6^\circ}$$

$$= \cot 3^\circ \Rightarrow \theta = 3^\circ$$

4. **Ans (C)**

Parametric equation of line through A(-1, 3)

$$\text{is } \frac{x+1}{\cos \theta} = \frac{y-3}{\sin \theta} = r$$

$\Rightarrow$ Any point on the line is

$$(r \cos \theta - 1, r \sin \theta + 3)$$

$$\therefore B(AB \cos \theta - 1, AB \sin \theta + 3)$$

$$\& C(AC \cos \theta - 1, AC \sin \theta + 3)$$

lies on $x + y = 4$ & $y = x$ respectively.

$$\Rightarrow AB \cos \theta - 1 + AB \sin \theta + 3 = 4$$

$$\text{or } AB(\cos \theta + \sin \theta) = 2 \quad \dots\dots\dots(1)$$

$$\& AC \cos \theta - 1 = AC \sin \theta + 3$$

$$\Rightarrow AC(\cos \theta - \sin \theta) = 4 \quad \dots\dots\dots(2)$$

$$\text{But } AB \cdot AC = 16$$

$$\therefore (1) \times (2) \Rightarrow 16(\cos 2\theta - \sin 2\theta) = 8$$

$$\text{or } \cos 2\theta = 1/2$$

$$\text{given } \tan \theta = \pm \frac{1}{\sqrt{3}}$$

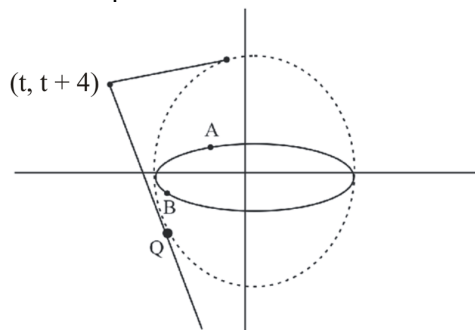
$$\therefore \left(\frac{m_1 - m_2}{1 + m_1 m_2} \right)^2 = \left(\frac{\frac{1}{\sqrt{3}} + \frac{1}{\sqrt{3}}}{1 - \frac{1}{3}} \right)^2 = 3$$

PART-3 : MATHEMATICS

SECTION-I(ii)

5. **Ans (B,D)**

$$\text{Ellipse } \frac{x^2}{4} + y^2 = 1$$



$$P = (x_1, y_1) \quad A = \left(x_1, \frac{b}{a}y_1\right)$$

$$Q = (x_2, y_2) \quad B = \left(x_2, \frac{b}{a}y_2\right)$$

$$A = \left(x_1, \frac{1}{2}y_1\right) \quad B = \left(x_2, \frac{1}{2}y_2\right)$$

Mid point of AB = (h, k)

$$h = \frac{x_1 + x_2}{2} \quad k = \frac{y_1 + y_2}{2}$$

Mid point of PQ

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) = (h, 2k)$$

Equation of PQ by mid point form & chord of contact & compare.

$$\text{Circle } x^2 + y^2 = 4$$

$$xt + y(t + 4) = 4 \quad \text{Chord of contact}$$

$$hx + 2ky = h^2 + 4k^2 \quad \text{Mid point chord}$$

$$\text{Compare } \frac{h}{t} = \frac{2k}{t+4} = \frac{h^2 + 4k^2}{4}$$

$$2k - h = h^2 + 4k^2$$

$$x^2 + 4y^2 - 2y + x = 0$$

$$\text{Ellipse with } e = \sqrt{1 - \frac{1}{4}} = \frac{\sqrt{3}}{2}$$

6. **Ans (B,C)**

Let equation be $x + 4y + c = 0$

Put in $x^2 + 4xy + 3y^2 - 5x - 6y + 3 = 0$ & $D = 0$

$$3x^2 + x(-C - x) +$$

$$3\left(\frac{-C-x}{4}\right)^2 - 5x - 6\left(\frac{-C-x}{4}\right) + 3 = 0$$

$$D = 0$$

$$C = -5, -8$$

Equation of T_1 & T_2 is $x + 4y - 5 = 0$ & $x + 4y - 8 = 0$.

7. **Ans (A,B,C)**

$$I_n = \int_0^\pi \frac{2 \cos x - \cos(2n-1)x - 2 \cos(2nx) - \cos(2n+x)x + 2}{1 - \cos 2x} dx$$

$$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$$

$$I_n = \int_0^\pi \frac{-2 \cos x + \cos(2n-1)x - 2 \cos(2nx) + \cos(2n-1)x + 2}{1 - \cos 2x} dx$$

$$2I_n = \int_0^\pi \frac{4(1 - \cos(2nx))}{1 - \cos 2x} dx$$

$$I_n = 2 \int_0^\pi \frac{\sin^2 nx}{\sin^2 x} dx$$

$$f(x) = f(a+b-x)$$

$$I_n = 4 \int_0^{\pi/2} \frac{\sin^2 nx}{\sin^2 x} dx$$

$$I_{n+1} - I_n = 4 \int_0^{\pi/2} \frac{\sin^2(n+1)x - \sin^2 x}{\sin x} dx$$

$$I_{n+1} - I_n = 4 \int_0^{\pi/2} \frac{\sin(2n+1)x}{\sin x} dx \quad \dots(1)$$

$$\text{Similarly } I_{n+2} - I_{n+1} = 4 \int_0^{\pi/2} \frac{\sin(2n+3)x}{\sin x} dx \quad \dots(2)$$

$$(1) \dots\dots\dots (2) = 0$$

$$I_{n+1} - I_n = I_{n+2} - I_{n+1}$$

$$2I_{n+1} = I_n + I_{n+2}$$

$I_0, I_1, I_2 \dots\dots\dots$ are in A.P.

$$I_0 = 4 \int_0^{\pi/2} 0 dx = 0 \quad I_1 = 4 \int_0^{\pi/2} \frac{\sin^2 x}{\sin^2 x} dx = 4 \frac{\pi}{2}$$

A.P. will be $I_0 = 0, I_1 = 2\pi, I_2 = 4\pi \dots\dots\dots$

$$\sum_{n=0}^{99} I_n = I_0 + I_1 + \dots\dots\dots I_{99} = \frac{100}{2} [0 + 198\pi] = 9900\pi$$

$$I_{1011} = 2022\pi$$

8. Ans (A,D)

$$\frac{2x}{(x-1)(x-4)} = \frac{-2/3}{(x-1)} + \frac{8/3}{x-4}$$

$$\int \frac{e^{x-1}}{(x-1)(x-4)} 2x dx = \int \frac{e^{x-1} \left(-\frac{2}{3}\right)}{(x-1)} + \frac{e^{x-4} \frac{8}{3}}{(x-4)} e^3 dx$$

9. Ans (B,D)

$$y + y' = y + y' + \dots \dots \infty$$

$$y = y' + \frac{d}{dx} (y' + y + y' \dots \infty)$$

$$y = y' + \frac{dy}{dx}$$

$$y = 2y$$

$$dx = \frac{2dy}{y}$$

$$x = 2 \ln y + C$$

$$y = e^{\frac{x}{2}-C}$$

$$f(0) = \frac{1}{e^{\frac{x}{2}-1}}$$

$$f(x) = e^{\frac{x}{2}-1}$$

$$y + y^i = y^{ii} + y^{iii} + y^{iv} \dots \dots \infty$$

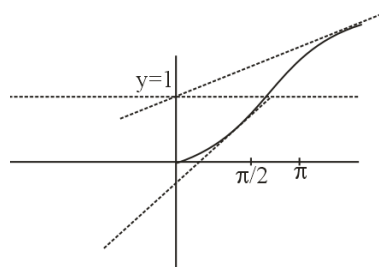
$$y + y^i = y^{ii} + \frac{d}{dx} (y^{ii} + y^{iii} \dots \dots \infty)$$

$$y + y^i = y^{ii} + \frac{d}{dx} (y + y^i)$$

$$y + y^i = y^{ii} + y^i + y^{ii}$$

$$y = 2y^{ii}$$

10. Ans (A,B)



$$f(x) = 1 - \cos x$$

$$g(x) = \frac{\sqrt{3}}{2} |x| + a$$

$$f'(x) = \sin x$$

$$g'(x) = \frac{\sqrt{3}}{2} (x > 0)$$

$$\sin x = \frac{\sqrt{3}}{2}$$

$$x = \frac{\pi}{3}, \frac{2\pi}{3}$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (2.00)

$$\vec{r} = \vec{a} + \vec{a} \times \hat{b}$$

$$\vec{r} \cdot \vec{a} = a^2 \Rightarrow \frac{\vec{r} \cdot \vec{a}}{a} = a = 2$$

$$\Rightarrow a = 2$$

2. Ans (8.00)

$$r^2 = a^2 + |\vec{a} \times \hat{b}|^2$$

$$= 4 + a^2 - (\vec{a} \cdot \vec{b})^2 = 8 - 4 \cos \theta$$

$$\Rightarrow r_{\max}^2 = 8$$

3. Ans (5.00)

$$L_1 : \frac{x-2}{-\frac{1}{2}} = \frac{y-2}{\frac{1}{2}} = \frac{z-4}{1}$$

$$\text{and } L_2 : \frac{x-0}{\frac{1}{2}} = \frac{y+1}{1} = \frac{z-3}{1}$$

$$b_1 \times b_2 = \vec{n} = 2\hat{i} + 0\hat{j} + \hat{k}$$

$$\text{Line will be } R = (-1, 0, 2) + \lambda(2, 0, 1)$$

Distance from origin will be

$$= \frac{5}{\sqrt{5}} = \sqrt{5}$$

4. Ans (6.00)

Using section formula and eliminating

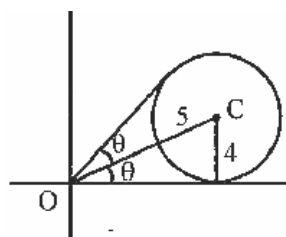
$$\text{we get } w + 2u = 6$$

$$\therefore \text{ locus of } M(u, v, w) \text{ is}$$

$$2x + z = 6$$

5. Ans (24.00)

$$|z - 3 - 4i| = 4$$



$$(x - 3)^2 + (y - 4)^2 = 4^2$$

z will lie on circle

given in figure

$$\text{amp}(z)|_{\min} = 0 = \theta_1$$

$$\text{amp}(z)|_{\max} = 2\theta = \theta_2$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta} = -\frac{24}{7}$$

$$\tan(\theta_1 + \theta_2) = \tan(2\theta + \theta) = \tan(3\theta) = -\frac{24}{7}$$

6. Ans (40.00)

area of triangle formed by z, iz & z + iz

$$\text{is } \frac{1}{2}|z|^2$$

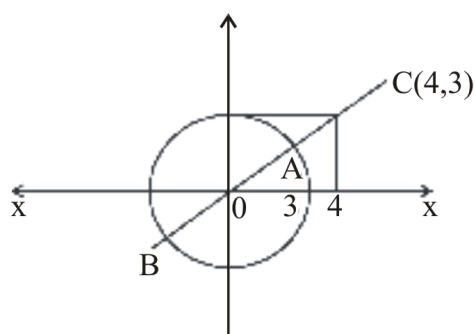
PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (9)

For two common tangents circles must be intersecting.

$$AC < \sqrt{16 + 9 - n^2} < BC$$



$$2 < \sqrt{25 - n^2} < 8 = 0 \leq n^2 \leq 16$$

$$\Rightarrow n = 0, \pm 1, \pm 2, \pm 3, \pm 4$$

$$= 9$$

8. Ans (8)

$$3[x] - 4x = 4 \quad \text{or} \quad 3[x] - 4x = -4$$

$$\{x\} = \frac{-4 - [x]}{4} \quad \{x\} = \frac{4 - [x]}{4}$$

$$-8 < [x] \leq -4$$

$$0 < [x] \leq 4$$

$$[x] = -4, -5, -6, -7$$

$$[x] = 1, 2, 3, 4$$

$$\{x\} = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$$

$$\{x\} = \frac{3}{4}, \frac{1}{2}, \frac{1}{4}, 0$$

4 solutions

4 solutions .

9. Ans (9)

$$P(A \cap \bar{B} \cap C) + P(A \cap \bar{B} \cap \bar{C}) = P(A \cap \bar{B}) = 1/7.$$

$$P(A \cap B) = P(A) - P(A \cap \bar{B}) = 1/7.$$

$$P(\bar{A}/B) = \frac{P(\bar{A} \cap B)}{P(B)} = \frac{1 - P(A \cup B)}{1 - P(B)} = \frac{4}{5}$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

FULL SYLLABUS

14-04-2024

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B,D	A,C	B,D	A,D	A,D	A,B,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	B	D	D	A		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	8.85	8.00	5.00	8.00	0.50	0.50
SECTION-II (ii)	Q.	7	8	9			
	A.	3	3	42			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B,D	C,D	C,D	A,D	A,C,D	A,C,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	B	A	C	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.00	3.33	5.00	8.00	3.30	6.01
SECTION-II (ii)	Q.	7	8	9			
	A.	81	4	06			

PART-3 : MATHEMATICS

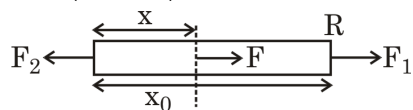
SECTION-I(i)	Q.	1	2	3	4	5	6
	A.	A,B,C,D	C,D	B,C	A,B,C	A,C	A,B,C
SECTION-I(ii)	Q.	7	8	9	10		
	A.	D	D	D	C		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.00	6.00	0.69	1.69	3.00	4.00
SECTION-II (ii)	Q.	7	8	9			
	A.	16	12	1			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,B,D)



$$F = \left(\frac{F_1 - F_2}{x_0} \right) x + F_2 = \alpha x + \beta$$

Energy density at any x

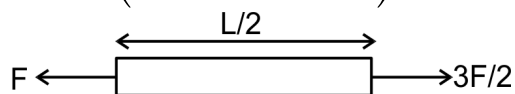
$$\frac{dU}{dV} = \frac{1}{2} \left(\frac{\alpha x + \beta}{A} \right) \left(\frac{\alpha x + \beta}{AY} \right) = \frac{1}{2A^2Y} (\alpha x + \beta)^2$$

Energy stored in small segment dx

$$dU = \frac{1}{2A^2Y} (\alpha^2 x^2 + \beta^2 + 2\alpha\beta x) A dx$$

$$U = \int dU = \frac{1}{2AY} \int_0^{x_0} (\alpha^2 x^2 + \beta^2 + 2\alpha\beta x) dx$$

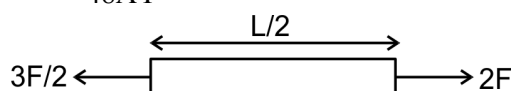
$$= \frac{1}{2AY} \left(\frac{\alpha^2 x_0^3}{3} + \beta^2 x_0 + \alpha\beta x_0^2 \right)$$



Consider section PQ

$$\alpha = F/L, \beta = F, x_0 = L/2$$

$$U_1 = \frac{19F^2L}{48AY}$$



Consider section QR

$$\alpha = F/L, \beta = 3F/2, x_0 = L/2$$

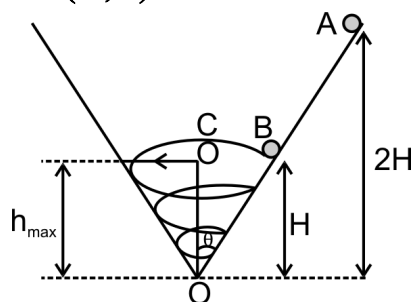
$$U_2 = \frac{37F^2L}{48AY}$$

Total elastic P.E

$$U = U_1 + U_2$$

$$U = \frac{7F^2L}{6AY}$$

2. Ans (A,C)



Collision elastic

$$\text{velocity after collision } V_B = \sqrt{2gH}$$

at B

Angular momentum about vertex 'O' $L_B = L_C$

$$m V_B H \cos \theta = m V_C h_{\max} \cos \theta$$

$$V_C = \frac{H}{h_{\max}} \sqrt{2gH}$$

ME conservation

$$KE_B + PE_B = KE_C + PE_C$$

$$\frac{1}{2} m V_B^2 + mgH = \frac{1}{2} m V_C^2 + mg h_{\max}$$

$$V_B^2 + 2gH = V_C^2 + 2gh_{\max}$$

$$4gH = \frac{H^2}{h_{\max}^2} 2gH + 2gh_{\max}$$

$$2 = \frac{1}{\left(\frac{h_{\max}}{H} \right)^2} + \frac{h_{\max}}{H} = \frac{1}{x^2} + x$$

$$\text{Solving } x = \frac{\sqrt{5} + 1}{2}$$

$$\therefore H_{\max} = \left(\frac{1 + \sqrt{5}}{2} \right) H$$

$$V = \sqrt{2gH \left(2 - \frac{1 + \sqrt{5}}{2} \right)} = \sqrt{gH(3 - \sqrt{5})}$$

$$= \sqrt{\frac{gH}{2} (\sqrt{5} - 1)^2} = \sqrt{2gH} \cdot \frac{\sqrt{5} - 1}{2}$$

$$= \frac{\sqrt{8gH}}{(\sqrt{5} + 1)}$$

3. Ans (B,D)

$$MX = m[R(1 - \cos\theta) - x]$$

$$\Rightarrow X = \frac{mR(1 - \cos\theta)}{(M + m)}$$

$$y_m = \frac{mR \sin\theta + M \times 0}{m + M} = \left(\frac{m}{M + m} \right) R \sin\theta$$

$$\frac{d\theta}{dt} = \frac{V}{R}$$

$$\frac{dx}{dt} = \frac{mR}{(m + M)} \sin\theta \frac{d\theta}{dt} = \frac{mV}{(m + M)} \sin\theta$$

$$\frac{d^2x}{dt^2} = \frac{mV}{(m + M)} (\cos\theta) \frac{d\theta}{dt}$$

$$= \frac{mV^2}{(m + M)R} \cos\theta$$

4. Ans (A,D)

Equation of continuity :

$$u \cdot \pi d^2 = V \cdot \pi \left(\frac{d}{2} \right)^2$$

$$\Rightarrow V = 4u = 4 \text{ m/s}$$

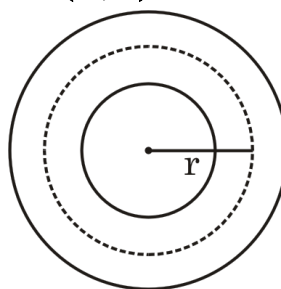
$$P_A + \frac{1}{2} \rho u^2 + 2000 = P_B + \frac{1}{2} \rho V^2$$

$$P_A - P_B = \frac{1}{2} \rho V^2 - \frac{1}{2} \rho u^2 - 2000$$

$$= \frac{1}{2} \times 1000(4^2 - 1^2) - 2000$$

$$= 500 \times 15 - 2000 = 5500 \text{ Pa}$$

5. Ans (A,D)



$$Q(r) = \int_{R_1}^r 4\pi r^2 dr \cdot (a + br)$$

$$= 4\pi \left[\frac{ar^3}{3} + b \frac{r^4}{4} \right]_{R_1}^r$$

$$= 4\pi \left[\frac{a}{3}(r^3 - R_1^3) + \frac{b}{4}(r^4 - R_1^4) \right]$$

$$E(r) \cdot 4\pi r^2 \cdot \epsilon_0 = 4\pi \left[\frac{a}{3}(r^3 - R_1^3) + \frac{b}{4}(r^4 - R_1^4) \right]$$

$$E(r) = \frac{1}{\epsilon_0 r^2} \left[\frac{a}{3}(r^3 - R_1^3) + \frac{b}{4}(r^4 - R_1^4) \right]$$

$$V(r) = \frac{1}{r} \int_{R_1}^r \frac{1}{4\pi \epsilon_0} \cdot 4\pi r^2 \cdot dr \cdot (a + br)$$

$$+ \int_r^{R_2} \frac{1}{4\pi \epsilon_0} \cdot \frac{4\pi r^2(a + br)}{r} dr$$

$$= \frac{1}{r \epsilon_0} \left[\frac{a}{3}(r^3 - R_1^3) + \frac{b}{4}(r^4 - R_1^4) \right]$$

$$+ \frac{1}{\epsilon_0} \left[\frac{a}{2}(R_2^2 - r^2) + \frac{b}{3}(R_2^3 - r^3) \right]$$

For $r < R_1$:

$$V(r) = \int_{R_1}^{R_2} \frac{1}{4\pi \epsilon_0} \cdot \frac{4\pi r^2 \cdot dr(a + br)}{r}$$

$$= \frac{1}{\epsilon_0} \left[\frac{a}{2}(R_2^2 - R_1^2) + \frac{b}{3}(R_2^3 - R_1^3) \right]$$

6. Ans (A, B, C)

$$Q_0 = \frac{V_0 R_1}{R_1 + R_2} \cdot C, Q_S \rightarrow \infty$$

$$i_0 = \frac{V_0}{R_1 + R_2}, \tau = R_1 C$$

$$2\pi r \cdot B = \mu_0 \cdot \frac{r^2}{b^2} \cdot \frac{V_0}{R_1 + R_2}$$

$$B = \frac{\mu_0 r V_0}{2\pi b^2 (R_1 + R_2)}; \quad \frac{dt}{R_1 \cdot \epsilon_0 \cdot \pi b^2}$$

$$S = \frac{1}{\mu_0} \cdot \left(\frac{\mu_0 r V_0}{2\pi b^2 (R_1 + R_2)} e^{-t/\tau} \right)$$

$$\left(\frac{V_0 R_1}{R_1 + R_2} \cdot C e^{-t/\tau} \right)$$

$$\epsilon_0 \cdot \pi b^2$$

$$= \frac{V_0^2 r R_1}{2\pi b^2 (R_1 + R_2)^2} \cdot \frac{\pi b^2}{d} \frac{e^{-2t/\tau}}{\pi b^2}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (B)

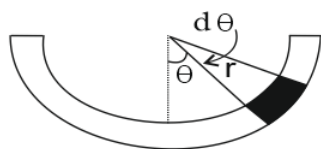
$$E = \frac{V}{L} \text{ and } J = \sigma E$$

$$J = \sigma E \cos \theta$$

So with increasing θ J decrease

8. Ans (D)

Current in the element



$$di = E \sigma_0 R t \cos \theta d\theta$$

$$di = j dA = j (R d\theta t);$$

$$di = \frac{V \sigma_0 R t \cos \theta d\theta}{L}$$

$$I = \left[\frac{V \sigma_0 R t}{L} \right] \int_{-\pi/2}^{\pi/2} \cos \theta d\theta = \frac{2 \sigma_0 V R t}{L}$$

10. Ans (A)

The energy provided by the power supply is converted into two aspects: the first is provided to the submarine, and the second is the heat of sea water energy,

The first part of the electrical power provided by the power source is the traction power

$$P_1 = (Fv) \times 12 = 12 (BIc) v_0 = 6.91 \times 10^5 W$$

The second part of the electrical power provided by the power supply is the Joule heat power of seawater per unit time.

Resistance of sea water in thruster

$$R = \frac{\rho c}{ab} = 0.5 \Omega$$

$$P_2 = (I^2 R) 12 = 10^6 \times 0.5 \times 12 = 6 \times 10^6 W$$

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (8.85)

$$Q = CV \Rightarrow ne = \frac{\epsilon_0 A}{d} V$$

$$n = \frac{8.85 \times 10^{-12} \times 50 \times 10^{-4}}{0.5 \times 10^{-3} \times 1.6 \times 10^{-19}} \times 16$$

$$n = 8.85 \times 10^9$$

2. Ans (8.00)

Stopping potential, $V_S = 8V$

and $KE = eV_S$

$$\therefore KE = 8 \text{ eV}$$

5. Ans (0.50)

$$2LI = (2L + 2M)I'$$

$$I' = \frac{2L}{2L + 2M} I = \frac{2}{4} I = \frac{1}{2} I$$

6. Ans (0.50)

$$= 2 \times \frac{1}{2} LI^2 + \frac{1}{2} MI^2 \times 2 - 2 \times \frac{1}{2} LI^2$$

$$= (L + M)I^2 - LI^2$$

$$= 2L \left(\frac{1}{2} I \right)^2 - LI^2 = -\frac{1}{2} LI^2$$

PART-1 : PHYSICS

SECTION-II (ii)

7. **Ans (3)**

Volume of platform towards

$$\text{left} = \frac{m \times \frac{9}{13}}{m + 2m} = \frac{3}{13} \text{ m/s}$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\Rightarrow \frac{1}{-f} = \frac{1}{v} + \frac{1}{-\frac{3f}{2}}$$

$$\frac{1}{v} = \frac{2}{3f} - \frac{1}{f} = \frac{-1}{3f}$$

$$v = -3f$$

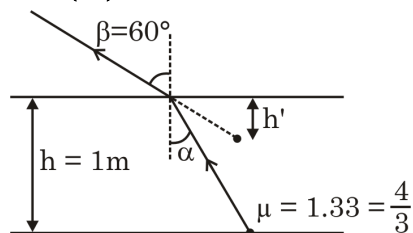
$$\frac{dv}{dt} = -\frac{du}{dt} \left(\frac{v^2}{u^2} \right)$$

$$= - \left(\frac{-9}{13} \right) \left(\frac{-3f}{-3f/2} \right)^2 = \frac{36}{13} \text{ m/s (w.r.t. mirror)}$$

$$\text{velocity of image} = \frac{36}{13} + \frac{3}{12} = 3 \text{ m/s}$$

(w.r.t. ground)

8. **Ans (3)**



$$1 \cdot \sin 60^\circ = \frac{4}{3} \sin \alpha$$

$$\sin \alpha = \frac{3}{4} \times \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{8}$$

$$\cos \alpha = \frac{\sqrt{37}}{8}; \cos \beta = \frac{1}{2}$$

$$\therefore h' = \frac{h \cos^3 \beta}{\mu \cos^3 \alpha}$$

$$= \frac{14}{\frac{4}{3}} \times \frac{\frac{1}{8}}{\frac{37\sqrt{37}}{8 \times 8 \times 8}} = 2.982 \text{ m}$$

9. **Ans (42)**

$$Q = 6.40 \times 10^{-12} \times 220 = \text{Ne}$$

$$N = \frac{6.40 \times 10^{-12} \times 220}{1.6 \times 10^{-19}} = 88 \times 10^8$$

$$= 8.8 \times 10^9$$

$$\text{Rate of decay} = \frac{dN}{dt} = \frac{8.8 \times 10^9}{88} = 10^8$$

$$\frac{dN}{dt} = \lambda N = \frac{\ln 2}{(T_{1/2})} N$$

$$10^8 = \frac{\ln 2}{(T_{1/2})} \times \frac{3.06 \times 10^{-3}}{306} \times \frac{6.023 \times 10^{23}}{100}$$

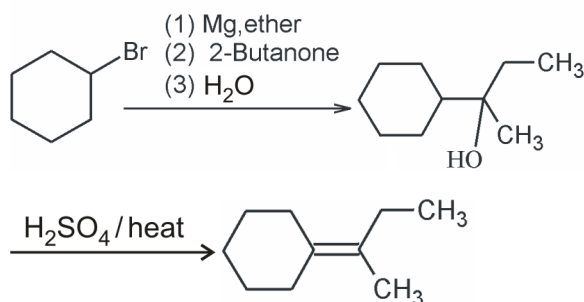
$$10^8 = \frac{0.7}{(T_{1/2})} \times 6.023 \times 10^{18}$$

$$T_{1/2} = 4.2 \times 10^{-10} = 42 \times 10^9 \text{ sec}$$

PART-2 : CHEMISTRY

SECTION-I (i)

3. **Ans (C,D)**



4. **Ans (A,D)**

Fact basis

6. **Ans (A,C,D)**

Zeolite is an example of 3D silicates.

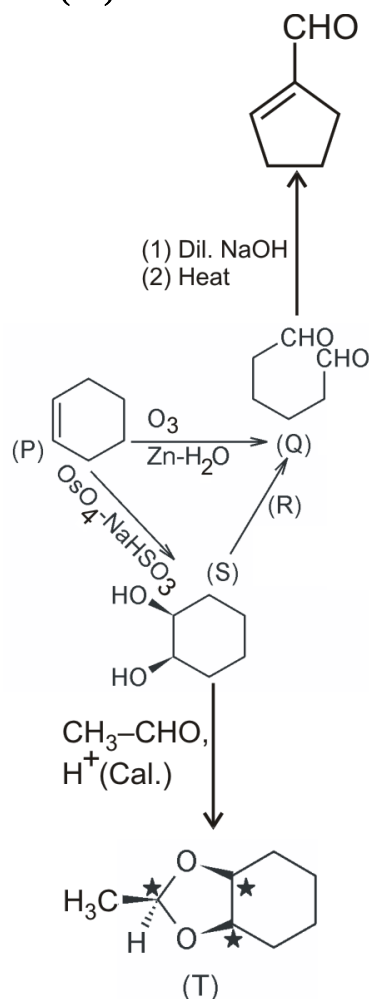
PART-2 : CHEMISTRY

SECTION-I (ii)

7. **Ans (B)**

CoCl₃.5NH₃ can be represented as [Co(NH₃)₅Cl] Cl₂

10. Ans (B)



PART-2 : CHEMISTRY

SECTION-II (i)

1. Ans (0.00)

Product is $\text{Na}_4[\text{Fe}(\text{CN})_5\text{NOS}]$ in both reactant & product Fe shown +2 oxidation state

2. Ans (3.33)

Ionic equation is –



For 3 moles of $\text{S}^{2-} \longrightarrow 1 \text{ mole } \text{Cr}_2\text{O}_7^{2-}$ is required

$\therefore$ for 10 mole of $\text{S}^{2-} \longrightarrow \frac{10}{3} = 3.33$ moles of $\text{Cr}_2\text{O}_7^{2-}$ will be required

3. Ans (5.00)

(P) = CH_3COOH (Q) = $\text{CH}_3\text{CH}_2\text{OH}$ (R) = $\text{CH}_3\text{CH}_2\text{Br}$

(S) = CH_3CHO (T) = $\text{H}_3\text{C}-\text{CH}_2-\text{CH}(\text{OH})-\text{CH}_3$

(U) = $\text{H}_3\text{C}-\text{CH}_2-\text{C}(\text{OH})(\text{CH}_3)-\text{CH}_3$

(V) = $\text{H}_3\text{C}-\text{CH}_2-\text{C}(=\text{O})-\text{CH}_3$

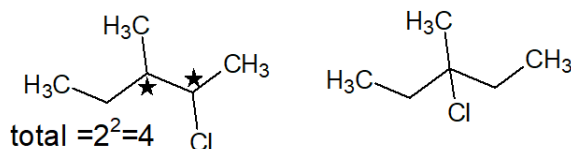
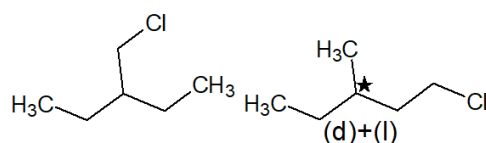
(W) = $\text{H}_3\text{C}-\text{CH}_2-\text{C}(\text{OH})(\text{CH}_3)-\text{CH}_3$

(Q), (R), (S), (T), (V) can give yellow ppt when react with NaOI

4. Ans (8.00)

(W) = $\text{H}_3\text{C}-\text{CH}_2-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{CH}_3$

Monochloro products =



6. Ans (6.01)

$$\text{Vol} = a^3 = (0.400)^3 = 6.4 \times 10^{-29} \text{ m}^3$$

$$\text{Mass} = \rho \times V = (2.0625 \times 10^4) \times (6.4 \times 10^{-29}) = 1.32 \times 10^{-24}$$

$\therefore$ Mass of one Au atom

$$= \frac{1.32 \times 10^{-24}}{4} = 3.3 \times 10^{-25} \text{ kg}$$

$$\therefore \text{Avogadro's constant} = \frac{198.33 \text{ gm/mol}}{3.3 \times 10^{-22} \text{ gm}} = 6.01 \times 10^{23}$$

PART-2 : CHEMISTRY

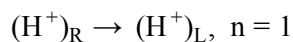
SECTION-II (ii)

7. Ans (81)

$$\Delta G = V \Delta P = 90 \times 10^{-3} \times 9 = 810 \times 10^{-3} \text{ lit-bar}$$

$$\Delta G = 81 \text{ Joule}$$

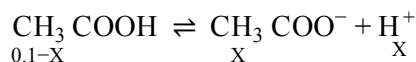
8. Ans (4)



$$0.12 = -\frac{0.06}{1} \log \frac{[H^+]_{\text{L}}}{[H^+]_{\text{R}}}$$

$$\Rightarrow \frac{[H^+]_{\text{L}}}{[H^+]_{\text{R}}} = 10^{-2}$$

R.H.S



$$\frac{X^2}{0.1 - X} = 10^{-5} \Rightarrow [H^+]_{\text{R}} = 10^{-3} \text{ M}$$

$$\text{then } [H^+]_{\text{L}} = 10^{-5} \text{ M}$$

L.H.S.



$$K_h = \frac{10^{-14}}{K_b} = \frac{x^2}{1-x} = \frac{(10^{-5})^2}{1-10^{-5}} = 10^{-10}$$

$$K_b = 10^{-4} \Rightarrow \text{p}K_b = 4$$

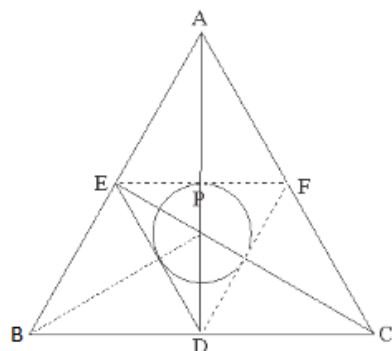
9. Ans (06)

Two stereoisomers will be optically inactive

PART-3 : MATHEMATICS

SECTION-I(i)

1. Ans (A,B,C,D)



P will be orthocentre of $\triangle ABC$ as altitudes AD and CE meet at P.

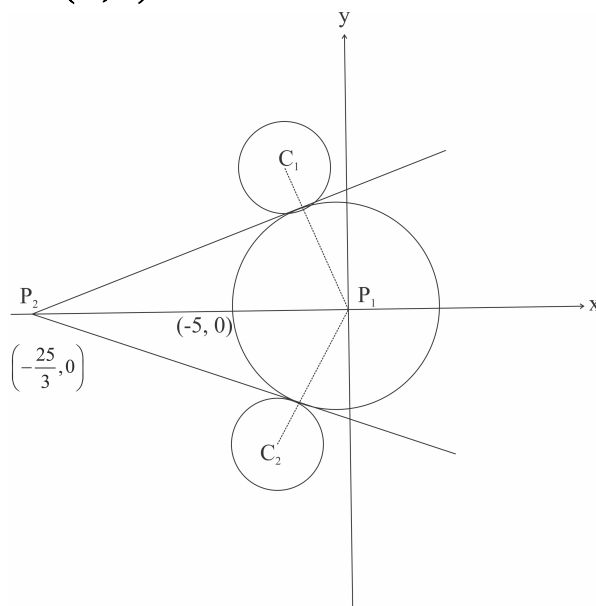
$\triangle DEF$ is the triangle formed by joining the foot of perpendicular. So P will be incentre of $\triangle DEF$.

$$\angle PDB = \angle PEB = 90^\circ$$

So PDCE is a cyclic quadrilateral.

$$\angle PBD = \angle PED = 90^\circ - C$$

2. Ans (C,D)



For Pair of tangent

$$y - 0 = M \left(x + \frac{25}{3} \right) \Rightarrow M = \pm \frac{3}{4}$$

$$\text{Equation of tangent } 3x - 4y \pm 25 = 0$$

$$T_1(-3, 4) \quad T_2(-3, -4)$$

$$\text{Using section formula } C_1 \left(-\frac{24}{5}, \frac{32}{5} \right) \\ C_2 \left(-\frac{24}{5}, -\frac{32}{5} \right)$$

$$\text{Area of } P_1C_1C_2 = \frac{1}{2} \times \frac{24}{5} \times \frac{64}{5} = \frac{768}{25}$$

$$\text{Area of } P_2C_1C_2 =$$

$$\frac{1}{2} \times \left(-\frac{24}{5} + \frac{25}{3} \right) \times \frac{64}{5} = \frac{1696}{75}$$

3. Ans (B,C)

$$30030 = 2 \times 3 \times 5 \times 7 \times 11 \times 13$$

$$\text{Divide in groups of } 5 \text{ \& } 1 \Rightarrow \frac{6!}{5!} = 6$$

$$3 \text{ \& } 3 \Rightarrow \frac{6!}{3!3!2!} = 10$$

$$2, 2 \text{ \& } 2 \Rightarrow \frac{6!}{2!2!2!3!} = 15$$

$$4, 1 \text{ \& } 1 \Rightarrow \frac{6!}{4!1!1!2!} = 15$$

$$4 \text{ \& } 2 \Rightarrow \frac{6!}{4!2!} = 15 \quad 3, 2 \text{ \& } 1 \Rightarrow \frac{6!}{3!2!1!} = 60$$

$$\text{For } a < b < c = k = 121$$

$$\text{Otherwise } = (121) \cdot 3 + 3 = 729$$

4. Ans (A,B,C)

M is mid point of S and P $\therefore M = (1, 4)$

$\therefore$ Equation of tangent at vertex is $x + y = 5$

= Equation of axis is $x - y = 3$

$$\therefore a = SO = 2\sqrt{2}$$

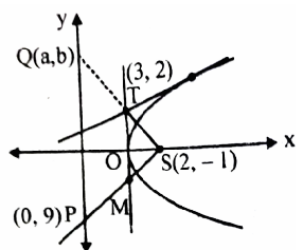


Image of S in tangent line is Q

$\therefore T$ is the mid point of S & Q. $\therefore 4a = 8\sqrt{2}$

5. Ans (A,C)

$f(n, x)$

$$\begin{aligned} &= \sum_{k=0}^n {}^nC_k \sin kx \cos(n-k)x = \frac{1}{2} \sum_{k=0}^n ({}^nC_k \sin nx + {}^nC_k \sin(2kx - nx)) \\ &= \frac{2^n}{2} \sin(nx) + \frac{1}{2} \sum_{k=0}^n {}^nC_k \sin(2kx - nx) = 2^{n-1} \sin nx + 0 \end{aligned}$$

$$f(n, x) = 2^{n-1} \sin(nx)$$

6. Ans (A,B,C)

$$2 \cot 2\theta = \cot \theta - \tan \theta$$

$$(2 \cot 4\theta = \cot 2\theta - \tan 2\theta) \times 2$$

$$(2 \cot 8\theta = \cot 4\theta - \tan 4\theta) \times 2^2$$

-

-

-

$$(2 \cot 2^n \theta = \cot(2^{n-1} \theta) - \tan 2^{n-1} \theta) \times 2^{n-1}$$

$$2^n \cot 2^n \theta = \cot \theta - [\tan \theta + 2 \tan 2\theta + 2^2 \tan 2^2 \theta + \dots + 2^{n-1} \tan 2^{n-1} \theta]$$

$$\sum_{r=1}^n 2^{r-1} \tan 2^{r-1} \theta = \cot \theta - 2^{2016} \theta$$

$$\text{So, } \sum_{r=1}^{2016} 2^{r-1} \tan 2^{r-1} \theta = \cot \theta - 2^{2016} \cot(2^{2016} \theta)$$

$$\theta = \frac{\pi}{2^{2018}} = \cot \frac{\pi}{2^{2018}} - 2^{2016} \cot 2^{2016} \frac{\pi}{2^{2018}} = \cot \frac{\pi}{2^{2018}} - 2^{2016}$$

$$\text{So, } \lambda = 2018, \mu = 2016$$

$$x^{\frac{1}{x}} \text{ is maximum } x = e, \text{ So}$$

$$(2016)^{1/2016} > (2018)^{1/2018}$$

PART-3 : MATHEMATICS

SECTION-I(ii)

7. Ans (D)

$$y = \left(1 + \frac{1}{x}\right)^x \quad \text{Domain } (-\infty, -1) \cup (0, \infty)$$

$$y \text{ is always +ive} \quad \log y = x \log \left(1 + \frac{1}{x}\right)$$

$$\frac{1}{y} \frac{dy}{dx} = \log \left(1 + \frac{1}{x}\right) + \frac{x}{1 + \frac{1}{x}} \left(-\frac{1}{x^2}\right)$$

$$\frac{dy}{dx} = y \left[\log \left(1 + \frac{1}{x}\right) - \frac{1}{1+x} \right]$$

$$g'(x) = \frac{-1}{(1+x)x} + \frac{1}{(1+x)^2} = -\frac{1}{x(1+x)^2}$$

$$g(0^+) = \infty$$

$$g(-1^-) = \text{increasing function}$$

$$g(\infty) = 0$$

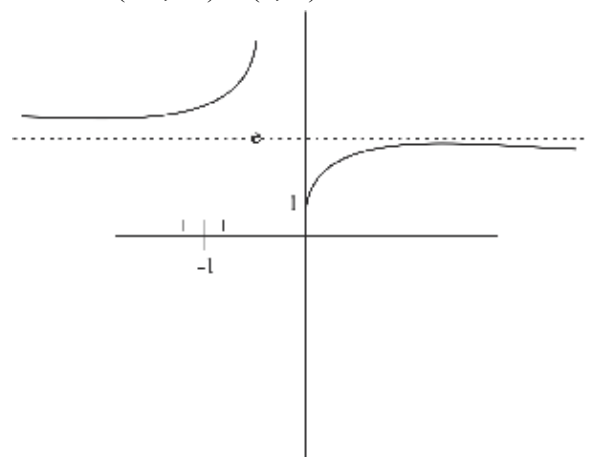
$$g(-\infty) = 0$$

So $g(x)$ is +ive

$$\frac{dy}{dx} \rightarrow \text{+ive}$$

$$f(x) = y = \left(1 + \frac{1}{x}\right)^x \text{ always increasing}$$

$$\text{Domain } (-\infty, -1) \cup (0, \infty)$$



$$f(\infty) = 1^\infty$$

$$\lim_{x \rightarrow \infty} \frac{1}{x} \cdot x$$

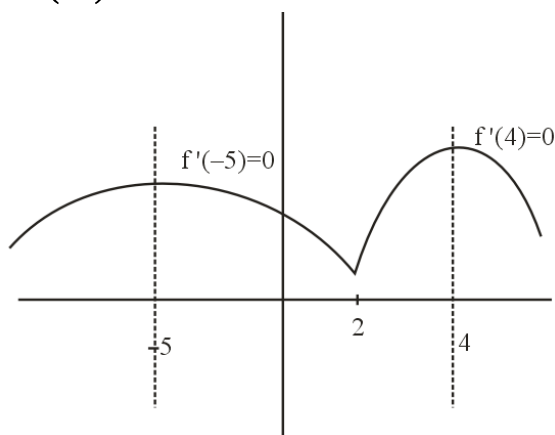
$$f(\infty) = e$$

$$f(0) = \infty^0$$

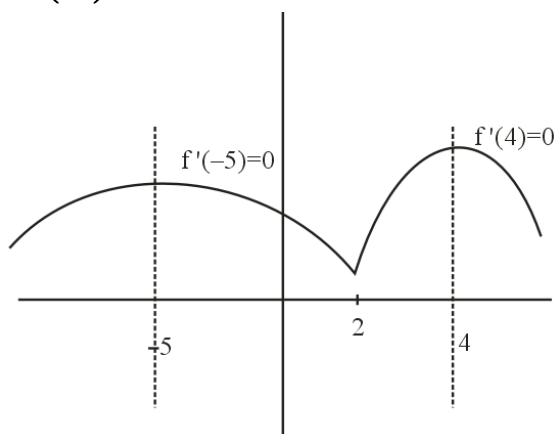
$$e^0 = 1$$

$$\lim_{x \rightarrow 0} \left(1 + \frac{1}{x}\right)^x = e^{\frac{\ln(1+\frac{1}{x})}{1/x}} \quad \begin{matrix} f(-1) \text{ by limit} \\ f(-\infty) \text{ by limit} \end{matrix}$$

9. Ans (D)



10. Ans (C)



PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (2.00)

Balls are identical box are identical

Just form groups 3 1 1 or 2 2 1 Ans. 2

2. Ans (6.00)

Balls are identical box will be considered different as they are in row so basically left middle right.

Distribution of identical objects each gets at least one

$${}^{5-1}C_{3-1} = {}^4C_2 = 6$$

3. Ans (0.69)

$$2 \times P(S \cap t) = 0.345 \times 2 = 0.69$$

4. Ans (1.69)

$$2 \times P(S \cup t) = 0.845 \times 2 = 1.69$$

5. Ans (3.00)

for $x < 0$

$$f(x) = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{e^x}{r(r+1)} = e^x$$

for $x > 0$

$$f(x) = ax \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{r^2 + r - 2}{r(r+1)} \right) + \lambda$$

$$= ax \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{\left(1 - 2 \left(\frac{1}{r} - \frac{1}{r+1} \right) \right)}{n} + \lambda$$

$$= ax \lim_{n \rightarrow \infty} \frac{n - 2 \left(1 - \frac{1}{n+1} \right)}{n} + \lambda = ax + \lambda$$

$$\Rightarrow f(x) = \begin{cases} e^x, & x < 0 \\ b, & x = 0 \\ ax + \lambda, & x > 0 \end{cases}$$

$\because f(x)$ is differentiable in $\mathbb{R}$

$$\Rightarrow a + b + \lambda = 3$$

$$f'(x) = \begin{cases} e^x, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

$$\Rightarrow S = f'(\log_e 2) + f'(\log_e 1/2) + f'(\log_e 3/22) + \dots$$

$$\infty = 1 + \frac{1}{2} + \frac{3}{2^2} + \frac{5}{2^3} + \dots \infty = 4$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (16)

$$\text{Note } |\text{adj } A| = |A|^{n-1}$$

$$|\text{adj } (\text{adj } A)| = A^{(n-1)^2}$$

8. Ans (12)

$$|z| = |z_1| = |z_2| = |z_3| = 2$$

$$\because |z_1 + z_2 + z_3|^2 \geq 0 \Rightarrow (z_1 + z_2 + z_3)(\bar{z}_1 + \bar{z}_2 + \bar{z}_3) \geq 0$$

$$\sum |z_1|^2 + \sum (z_1 \bar{z}_1 + \bar{z}_1 z_2) \geq 0$$

$$\Rightarrow 2 \sum |z_1|^2 + \sum (z_1 \bar{z}_2 + \bar{z}_1 z_2) \geq \sum |z_1|^2$$

$$\Rightarrow |z_1 + z_2|^2 + |z_2 + z_3|^2 + |z_3 + z_1|^2 \geq 12$$

9. Ans (1)

$$f(x) f(y) = f(x) + f(y) + f(xy) - 2$$

$$\text{Put } x = 1, y = 3$$

$$\Rightarrow f(1) = 2$$

$$\text{For } xy = 1$$

$$\Rightarrow y = \frac{1}{x}$$

$$\Rightarrow f(x) f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right) \Rightarrow f(x) = 1 \pm x^2$$

$$\therefore f(3) = 10 \Rightarrow f(x) = 1 + x^2$$